

Farey-Recursive Shortest Completions and Universal Spectra in Rational Balance Languages

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Abstract

Let p, q be coprime positive integers and, for $r \in \mathbb{Z}$, let

$$\mu_{p,q}(r) = \arg \min_{x,y \in \mathbb{Z}_{\geq 0}} \{x + y : px - qy = r\}$$

denote the shortest nonnegative integer completion vector associated with residual r . The problem is extracted from an explicit completion-aware 2:1 deterministic pushdown automaton and its $p:q$ generalization: the machine state and stack encode the demand still required to restore weighted balance. We study the arithmetic structure of the resulting ordered completion profiles for the rational balance language

$$\mathcal{L}_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}.$$

The principal completion-specific result is vector-valued. Writing a proper coprime pair as the Farey mediant of its oriented parents,

$$(p, q) = (c, u) + (v, d), \quad uv - cd = 1,$$

we prove a cross-parameter ordered Farey recursion: the entire residual-indexed completion profile of the mediant is partitioned by explicit complementary floor/ceiling index maps into an unshifted parent completion profile and a fixed translate of the other parent profile. Locally, consecutive positive completion vectors differ only by (u, c) or $(-d, -v)$; these two kernel-equivalent unit lifts provide the mechanism behind the recursion and allow the oriented parents to be recovered from the ordered profile. After a unimodular change of coordinates, this local two-step walk is a classical rational mechanical/Christoffel path, so the novelty claim is placed on the cross-parameter minimizer recursion rather than on the mechanical coding itself.

Scalar projection yields universal completion spectra. If $m = p + q$, then the combined positive and negative fundamental spectra have multiset

$$\{0, 0, 2, 3, \dots, m - 1\},$$

and every higher residual shell is its affine translate. Classical rational Beatty, mechanical/Christoffel, and Dedekind structures then arise as arithmetic projections of the vector theory. Computational audits are reported only as independent consistency checks; all mathematical results are proved symbolically.

Keywords. rational balance languages; shortest completion; linear Diophantine equations; Farey parents; Stern–Brocot tree; Beatty sequences; Christoffel words; Dedekind sums.

1 Introduction

For coprime positive integers p and q , consider the elementary-looking optimization problem

$$px - qy = r, \quad x, y \in \mathbb{Z}_{\geq 0}, \quad (1)$$

in which $x + y$ is minimized. For each residual $r \in \mathbb{Z}$, the minimizing pair is unique; we denote it by $\mu_{p,q}(r)$ and write

$$L_{p,q}(r) = |\mu_{p,q}(r)|_1.$$

The problem arises naturally from shortest right completion in the rational balance language

$$\mathcal{L}_{p,q} = \{w \in \{0, 1\}^* : qN_1(w) = pN_0(w)\},$$

For a word w , put $\rho(w) = qN_1(w) - pN_0(w)$. A right suffix containing x zeros and y ones balances w exactly when $px - qy = \rho(w)$, so the two coordinates of a completion vector record the required symbol counts. This completion interpretation arose from the author's completion-aware 2:1 DPDA and its later $p:q$ normalization theory [1]. We therefore retain that 2:1 machine explicitly in [section 2](#) as the operational provenance of the present arithmetic problem. All Farey, spectral, and moment proofs that follow are nevertheless self-contained and do not depend on the automaton construction.

At first sight, each residual r gives an isolated two-variable minimization problem. The main point of this paper is that the ordered family

$$(\mu_{p,q}(r))_{r \in \mathbb{Z}}$$

is highly structured. Its local evolution, its recursive organization across rational slopes, and its scalar spectra are governed by the unimodular Farey decomposition of q/p .

For a proper pair $p, q > 1$, let

$$(p, q) = (c, u) + (v, d) \quad (2)$$

be the oriented Farey-parent decomposition, normalized by

$$pu - qc = 1, \quad qv - pd = 1. \quad (3)$$

Thus $p = c + v$, $q = u + d$, and $uv - cd = 1$. The local mechanism is especially simple: consecutive positive shortest-completion vectors are obtained by choosing canonically between

$$(u, c), \quad (-d, -v).$$

Both raise the residual by one and their difference is the primitive homogeneous kernel (q, p) . Shortest completion therefore evolves by a unit-residual lift followed, when necessary, by canonical kernel reduction. The same two local jumps recover the two oriented Farey parents.

This local law is closely related to classical digital-line and Christoffel geometry: after a unimodular change of coordinates the completion walk is a rational mechanical path. Our principal completion-specific result is instead global and cross-parametric. The entire ordered profile at a Farey mediant is reconstructed from the canonical completion profiles of both parents by explicit complementary floor/ceiling index maps. One parent profile is inherited unchanged, while a translated copy of the second is inserted at interleaved residual positions. Iterating the construction decorates the Stern–Brocot tree by a recursively coherent family of shortest-completion profiles.

Scalar projection reveals a different phenomenon. If $m = p + q$, then the combined positive and negative fundamental completion-length spectra satisfy

$$\{L(0), \dots, L(p-1)\} \uplus \{L(0), L(-1), \dots, L(-(q-1))\} = \{0, 0, 2, 3, \dots, m-1\}.$$

More generally, every higher residual shell is an affine translate of this same multiset. Thus the vector geometry depends on the ratio $p : q$, whereas the combined scalar spectrum depends only on $p + q$.

Novelty boundary. The surrounding arithmetic objects have substantial classical theories. Rational Beatty sets and their Fourier structure are well established [5]; Christoffel words, standard factorization, continued fractions, and the Stern–Brocot tree are classical [2]. Christoffel duality already realizes modular inverses of numerator and denominator modulo the word length [3]. Arithmetic digital straight lines are an especially close neighbor: their standard lines are Diophantine strips, recognition uses parameters minimizing $a + b$, weakly exterior points are detected by remainders, and slope updates involve smallest positive Bézout coefficients and Stern–Brocot moves [7, 4, 8]. Likewise, Apéry sets in numerical semigroups select least representatives in residue classes [9], while minimal solutions of homogeneous linear Diophantine equations have been studied with notions of minimality different from the fixed-residual ℓ^1 problem used here [10]. Dedekind reciprocity is also classical [6].

Accordingly, we do not claim novelty for Diophantine-strip reduction, Bézout/Farey unit steps, modular representatives or inverses, mechanical/Christoffel coding, standard Christoffel factorization, digital-line Stern–Brocot updates, Beatty partitions, or Dedekind reciprocity in isolation. Indeed, [section 8](#) makes explicit that the local positive completion walk is unimodularly equivalent to a classical rational mechanical path. The principal completion-specific statement is the cross-parameter ordered recursion of [theorem 5.3](#): it compares the canonical fixed-residual ℓ^1 -minimizer sections for a Farey mediant and both of its parents, and identifies two interleaved residual-index subsequences of the child section with an unshifted parent section and a fixed translate of the other. We have not found this vector-labeled cross-parameter identity in the digital-line or Christoffel literature surveyed above; this is a cautious literature assessment rather than a priority claim.

Main contributions. The results are organized around one principal cross-parameter theorem and its local and scalar consequences:

- (i) **Cross-parameter Ordered Farey Recursion.** The full residual-indexed completion profile of a mediant is reconstructed from the two parent completion profiles through explicit complementary floor/ceiling index maps; the two parent blocks occur as interleaved subsequences, one after a fixed translation.
- (ii) **Local mechanism and inverse reading.** The canonical unit-lift law gives the two local completion increments and, conversely, those increments recover the two oriented Farey parents. These statements explain the local mechanism of the recursion; their mechanical/Christoffel shadow is classical after a unimodular coordinate change.
- (iii) **Recursive coherence.** Iterating the ordered parent-to-mediante construction produces a single recursively generated completion family over the Stern–Brocot tree, with the inherited and translated blocks specified explicitly at each step.
- (iv) **Universal scalar spectra.** The combined fundamental spectrum and all translated residual shells depend only on $m = p + q$.
- (v) **Classical arithmetic interfaces.** Rational Beatty complementarity, mechanical/Christoffel coding, digital-line Bézout updates, reflection symmetry, and Dedekind moments appear naturally after projection or change of coordinates.

2 The motivating completion-aware 2:1 DPDA

The arithmetic problem studied in this paper did not begin as an isolated Diophantine optimization. Its starting point was the completion-aware 2:1 deterministic pushdown design shown in [figure 1](#),

developed before the general $p:q$ theory and analyzed in detail in the companion automata-theoretic manuscript [1]. For

$$\mathcal{L}_{2,1} = \{w : N_1(w) = 2N_0(w)\},$$

the center state C , side states L, R , and binary work stack jointly encode what must still be supplied in order to complete the processed prefix to the language. Thus the relevant semantic object is not merely acceptance, but a shortest completion demand.

A transition label

$$a, A/\gamma$$

means that input a is consumed and stack top A is replaced by γ ; the leftmost work symbol is the stack top and Z_0 is the bottom marker. The dashed callout boxes in the figure are *not* ordinary ε -moves. They list completion operations enabled only when the right endmarker $\dashv$ is the next unread symbol. This separates ordinary binary-input processing from post-input completion.

Projecting away the initialization and acceptance wrapper leaves the semantic core $\{C, L, R\}$. If the physical stack is SZ_0 , define

$$D(S) = 2N_0(S) - N_1(S), \quad \Phi(C) = 0, \quad \Phi(L) = 3, \quad \Phi(R) = -3.$$

For every processed prefix the original machine satisfies the residual invariant

$$\rho_{2,1} = D(S) + \Phi(q_{\text{state}}), \quad \rho_{2,1} = N_1 - 2N_0. \quad (4)$$

Thus finite control may carry part of the completion demand even when the logical stack is empty. This is precisely why the arithmetic projection is useful: after forgetting how the demand is physically distributed between state and stack, only the residual and its unique shortest nonnegative completion remain.

Example 2.1 (The prefix 111). Ignoring Z_0 in the logical stack notation, the motivating machine follows

$$(C, 111 \dashv, \varepsilon) \vdash (C, 11 \dashv, 10) \vdash (C, 1 \dashv, 0) \vdash (L, \dashv, \varepsilon).$$

The processed prefix has residual $\rho_{2,1}(111) = 3$. The right-endmarker rule $\dashv, Z_0/100Z_0$ materializes the remaining state contribution and yields the completion word 100. If that suffix contains x zeros and y ones, balance requires

$$2x - y = 3,$$

and the unique minimum nonnegative solution is $(x, y) = (2, 1)$. In the notation of the present paper, this is exactly

$$\mu_{2,1}(3) = (2, 1).$$

The figure is included here as motivation and intellectual provenance, not as a proof dependency. The companion work studies the real-time $p:q$ implementation, WRITE-SEARCH-CANCEL normalization, and finite-control width [1]. The present paper begins after the operational projection: it takes the canonical map $r \mapsto \mu_{p,q}(r)$ as its object and studies how these minimizer profiles vary arithmetically across Farey-related ratios.

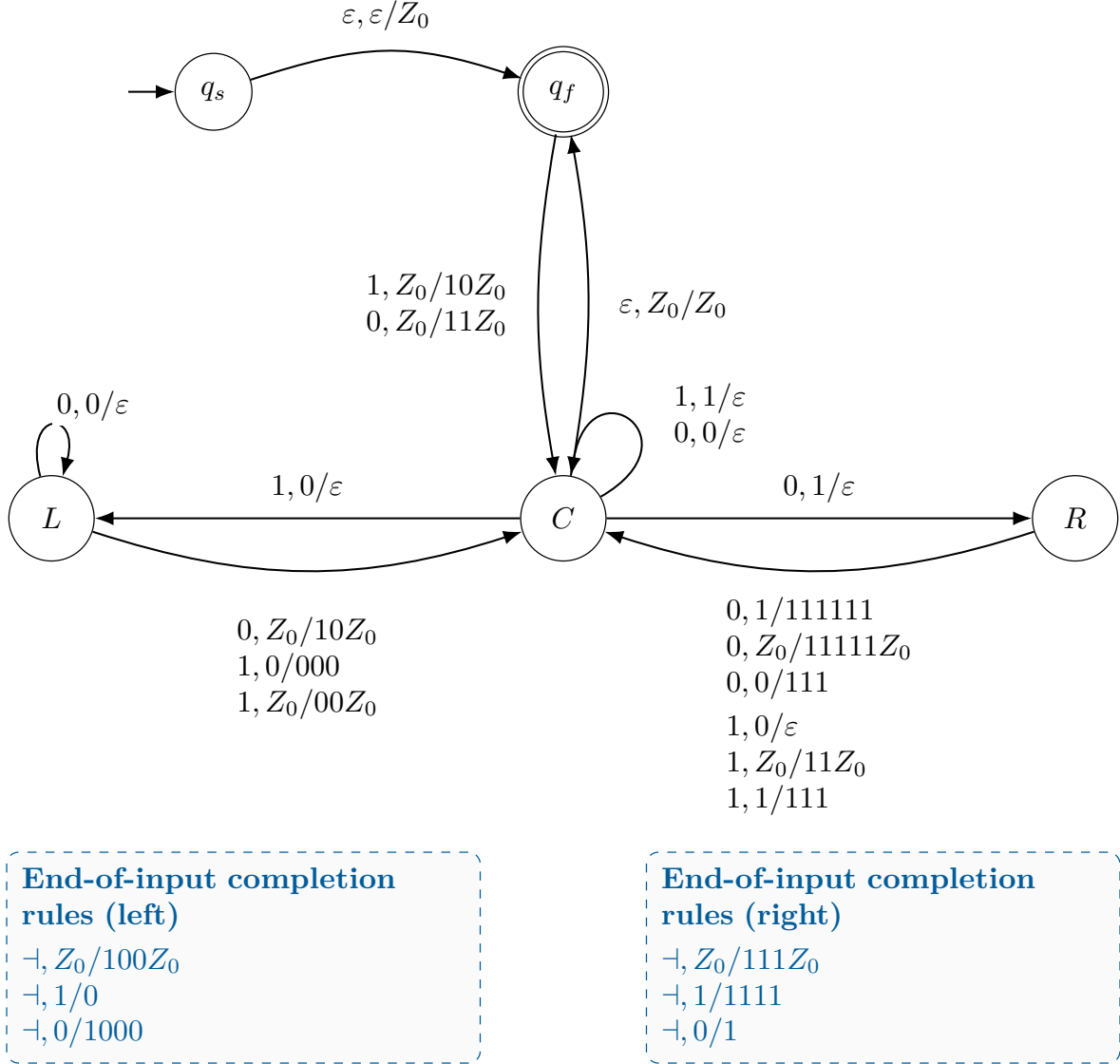


Figure 1: The original completion-aware 2:1 DPDA, in the empty-word-included presentation. Solid arcs process binary input; dashed boxes list only right-endmarker completion operations. This exact center/left/right construction is reproduced from the companion $p:q$ DPDA treatment [1] to make the provenance of the shortest-completion problem explicit.

3 Shortest-completion profiles and canonical strips

Fix coprime positive integers p, q . Define

$$W_{p,q}(x, y) = px - qy.$$

The primitive positive kernel vector of $W_{p,q}$ is

$$K_{p,q} = (q, p).$$

Thus any two integer solutions of $px - qy = r$ differ by an integral multiple of (q, p) .

Definition 3.1 (Shortest completion). For $r \in \mathbb{Z}$, define

$$\mu_{p,q}(r) = (x_r, y_r) = \arg \min_{x,y \in \mathbb{Z}_{\geq 0}} \{x + y : px - qy = r\},$$

and set

$$L_{p,q}(r) = x_r + y_r.$$

When p, q are fixed, we abbreviate these as $\mu(r)$ and $L(r)$.

Lemma 3.2 (Canonical-strip characterization). *For $r \geq 0$, $\mu(r)$ is the unique nonnegative solution of*

$$px - qy = r$$

satisfying

$$0 \leq y < p.$$

For $r \leq 0$, $\mu(r)$ is the unique nonnegative solution satisfying

$$0 \leq x < q.$$

Proof. Suppose $r \geq 0$. Since $\gcd(p, q) = 1$, the congruence

$$qy \equiv -r \pmod{p}$$

has a unique solution $y_0 \in \{0, \dots, p-1\}$. Put

$$x_0 = \frac{r + qy_0}{p} \geq 0.$$

Every integer solution is

$$(x_0, y_0) + k(q, p), \quad k \in \mathbb{Z}.$$

Because $0 \leq y_0 < p$, nonnegativity forces $k \geq 0$. Every other nonnegative solution therefore has length

$$x_0 + y_0 + k(p + q),$$

which is strictly larger for $k > 0$. Hence (x_0, y_0) is the unique minimizer. The case $r \leq 0$ is dual, using the unique representative $x_0 \in \{0, \dots, q-1\}$. $\square$

Corollary 3.3 (Kernel reduction). *If (x, y) is any nonnegative solution of residual r , then*

$$(x, y) = \mu(r) + k(q, p)$$

for a unique $k \geq 0$, and

$$x + y = L(r) + k(p + q).$$

Proposition 3.4 (Boundary families). *The cases $p = 1$ or $q = 1$ are explicit. If $p = 1$, then*

$$\mu_{1,q}(0) = (0, 0), \quad \mu_{1,q}(1) = (1, 0),$$

and for $1 \leq s \leq q$,

$$\mu_{1,q}(-s) = (q - s, 1).$$

If $q = 1$, then

$$\mu_{p,1}(0) = (0, 0), \quad \mu_{p,1}(-1) = (0, 1),$$

and for $1 \leq r \leq p$,

$$\mu_{p,1}(r) = (1, p - r).$$

Proof. Each displayed pair solves the relevant equation and lies in the canonical strip of [theorem 3.2](#). $\square$

4 Canonical Farey unit-lift dynamics

Assume throughout this section that $p, q > 1$. Let

$$(p, q) = (c, u) + (v, d) \quad (5)$$

be the oriented Farey-parent decomposition characterized by

$$pu - qc = 1, \quad qv - pd = 1. \quad (6)$$

It follows that

$$uv - cd = 1. \quad (7)$$

The existence and uniqueness of this oriented parent pair are classical Farey/Stern–Brocot facts; see, for example, [2].

Theorem 4.1 (Canonical Farey Unit-Lift Theorem). *Define*

$$a = (u, c), \quad b = (-d, -v).$$

Then

$$W_{p,q}(a) = W_{p,q}(b) = 1$$

and

$$a - b = (q, p) = K_{p,q}.$$

If

$$\mu(r) = (x_r, y_r), \quad 0 \leq r < p,$$

then

$$\mu(r+1) = \begin{cases} \mu(r) + (u, c), & 0 \leq y_r < v, \\ \mu(r) + (-d, -v), & v \leq y_r < p. \end{cases} \quad (8)$$

Proof. By [equation \(6\)](#),

$$W_{p,q}(u, c) = pu - qc = 1$$

and

$$W_{p,q}(-d, -v) = -pd + qv = 1.$$

Moreover,

$$(u, c) - (-d, -v) = (u + d, c + v) = (q, p).$$

Thus the two vectors are kernel-equivalent lifts of residual increment $+1$.

If $y_r < v$, then

$$0 \leq y_r + c < v + c = p,$$

so $\mu(r) + (u, c)$ has residual $r + 1$ and remains in the positive canonical strip. By [theorem 3.2](#), it is $\mu(r + 1)$.

If $y_r \geq v$, then adding (u, c) leaves the strip because $y_r + c \geq p$. Subtracting one kernel gives

$$(u, c) - (q, p) = (-d, -v),$$

and the new second coordinate is $y_r - v \in [0, p - 1]$. Hence $\mu(r) + (-d, -v)$ is the unique canonical representative of residual $r + 1$. $\square$

Remark 4.2 (Canonical unit lifting). The mechanism in [theorem 4.1](#) is intrinsic to shortest completion:

$$\text{unit residual lift} \quad + \quad \text{canonical kernel reduction}.$$

The Farey parent vectors appear as the two kernel-equivalent lifts available to this reduction.

Theorem 4.3 (Negative unit-lift theorem). *Define*

$$a^- = (d, v), \quad b^- = (-u, -c).$$

Then

$$W_{p,q}(a^-) = W_{p,q}(b^-) = -1, \quad a^- - b^- = (q, p).$$

If

$$\mu(-s) = (x_s, y_s), \quad 0 \leq s < q,$$

then

$$\mu(-(s+1)) = \begin{cases} \mu(-s) + (d, v), & 0 \leq x_s < u, \\ \mu(-s) + (-u, -c), & u \leq x_s < q. \end{cases} \quad (9)$$

Proof. The two increments have residual -1 because

$$pd - qv = -1, \quad -pu + qc = -1,$$

and their difference is (q, p) . If $x_s < u$, adding (d, v) keeps the first coordinate below $u + d = q$. If $x_s \geq u$, this lift leaves the negative canonical strip, and subtracting the kernel replaces (d, v) by $(-u, -c)$. The conclusion follows from [theorem 3.2](#). $\square$

Corollary 4.4 (Closed vector profiles). *For $0 \leq r \leq p$,*

$$\mu_{p,q}(r) = r(u, c) - \left\lfloor \frac{cr}{p} \right\rfloor (q, p). \quad (10)$$

For $0 \leq s \leq q$,

$$\mu_{p,q}(-s) = s(d, v) - \left\lfloor \frac{ds}{q} \right\rfloor (q, p). \quad (11)$$

In particular,

$$\mu(p) = (1, 0), \quad \mu(-q) = (0, 1).$$

Proof. In the positive walk, the kernel correction occurs exactly when the second coordinate wraps under addition of c modulo p . The number of wraps among the first r steps is $\lfloor cr/p \rfloor$, giving [equation \(10\)](#). The negative formula is identical with d modulo q . At $r = p$ and $s = q$, use $pu - qc = 1$ and $qv - pd = 1$. $\square$

Example 4.5 (The pair 7:4). For $(p, q) = (7, 4)$ the oriented parents are $(c, u) = (5, 3)$ and $(v, d) = (2, 1)$. Hence the two positive unit lifts are $(3, 5)$ and $(-1, -2)$. The positive fundamental profile is

$$(0, 0), (3, 5), (2, 3), (1, 1), (4, 6), (3, 4), (2, 2), (1, 0),$$

whose successive increments are

$$(3, 5), (-1, -2), (-1, -2), (3, 5), (-1, -2), (-1, -2), (-1, -2).$$

The ordered recursion places the left-parent profile at indices

$$\left\lceil \frac{7r}{5} \right\rceil \in \{0, 2, 3, 5, 6, 7\}$$

and the translated right-parent block at indices

$$\left\lfloor \frac{7s}{2} \right\rfloor + 1 \in \{1, 4\}.$$

Thus the local two-jump law and the global floor/ceiling shuffle are visible in the same small example.

5 Local Farey recovery and ordered parent recursion

5.1 Local encoding and parent recovery

Theorem 5.1 (Vector jump law). *The positive fundamental profile has exactly two local increments,*

$$\mu(r+1) - \mu(r) \in \{(u, c), (-d, -v)\}, \quad 0 \leq r < p.$$

The jump (u, c) occurs v times and the jump $(-d, -v)$ occurs c times.

The negative fundamental profile has exactly the two increments

$$(d, v), \quad (-u, -c),$$

with multiplicities u and d , respectively.

Proof. The two-value statements are [theorems 4.1](#) and [4.3](#). In the positive profile, the number of kernel corrections is

$$\sum_{r=0}^{p-1} \left(\left\lfloor \frac{c(r+1)}{p} \right\rfloor - \left\lfloor \frac{cr}{p} \right\rfloor \right) = c.$$

Since there are $p = c + v$ jumps in total, the other jump occurs v times. The negative case is identical. $\square$

Corollary 5.2 (Parent recovery). *For $p, q > 1$, the ordered positive completion profile uniquely determines the oriented Farey parents. If its two local increments are*

$$(a, b), \quad (-e, -f),$$

then

$$(c, u) = (b, a), \quad (v, d) = (f, e),$$

and consequently

$$p = b + f, \quad q = a + e.$$

Proof. By [theorem 5.1](#), $(a, b) = (u, c)$ and $(-e, -f) = (-d, -v)$. $\square$

5.2 Ordered parent-to-median recursion

Write

$$\mathcal{M}_{p,q}^+ = (\mu_{p,q}(0), \mu_{p,q}(1), \dots, \mu_{p,q}(p))$$

and

$$\mathcal{M}_{p,q}^- = (\mu_{p,q}(0), \mu_{p,q}(-1), \dots, \mu_{p,q}(-q)).$$

Theorem 5.3 (Cross-parameter Ordered Farey Recursion). *Let $(p, q) = (c, u) + (v, d)$ be as in equations (5) and (6). For $0 \leq r \leq c$,*

$$\mu_{p,q}\left(\left\lceil \frac{pr}{c} \right\rceil\right) = \mu_{c,u}(r). \quad (12)$$

For $0 \leq s < v$,

$$\mu_{p,q}\left(\left\lfloor \frac{ps}{v} \right\rfloor + 1\right) = (u, c) + \mu_{v,d}(s). \quad (13)$$

Dually, for $0 \leq r \leq u$,

$$\mu_{p,q}\left(-\left\lfloor \frac{qr}{u} \right\rfloor\right) = \mu_{c,u}(-r), \quad (14)$$

and for $1 \leq s \leq d$,

$$\mu_{p,q}\left(-\left(\left\lfloor \frac{qs}{d} \right\rfloor - 1\right)\right) = (u, c) + \mu_{v,d}(-s). \quad (15)$$

Moreover, the positive index families

$$I_L^+ = \left\{ \left\lceil \frac{pr}{c} \right\rceil : 0 \leq r \leq c \right\}, \quad I_R^+ = \left\{ \left\lfloor \frac{ps}{v} \right\rfloor + 1 : 0 \leq s < v \right\}$$

form a disjoint partition of $\{0, 1, \dots, p\}$. The corresponding negative index families in equations (14) and (15) form a disjoint partition of $\{0, 1, \dots, q\}$.

Proof. For equation (12), let $\mu_{c,u}(r) = (x, y)$. Then

$$cx - uy = r, \quad 0 \leq y < c.$$

Set $R = px - qy$. Since $pu = qc + 1$,

$$\frac{pr}{c} = px - \frac{pu}{c}y = px - qy - \frac{y}{c} = R - \frac{y}{c}.$$

As $0 \leq y/c < 1$, we get

$$R = \left\lceil \frac{pr}{c} \right\rceil.$$

Since $y < c < p$, the same vector lies in the child canonical strip, proving equation (12).

For equation (13), let $\mu_{v,d}(s) = (x, y)$, so

$$vx - dy = s, \quad 0 \leq y < v.$$

Since $pd = qv - 1$,

$$\frac{ps}{v} = px - \frac{pd}{v}y = px - qy + \frac{y}{v},$$

and therefore

$$px - qy = \left\lfloor \frac{ps}{v} \right\rfloor.$$

Adding (u, c) raises the child residual by one. Moreover $0 \leq c + y < c + v = p$, so the translated vector is canonical. This proves [equation \(13\)](#).

For [equation \(14\)](#), let $\mu_{c,u}(-r) = (x, y)$, so $cx - uy = -r$ and $0 \leq x < u$. Then

$$\frac{qr}{u} = qy - \frac{qc}{u}x = qy - px + \frac{x}{u} = -(px - qy) + \frac{x}{u}.$$

Hence $px - qy = -\lfloor qr/u \rfloor$, and the same vector lies in the child negative strip because $x < u < q$.

For [equation \(15\)](#), let $\mu_{v,d}(-s) = (x, y)$, so $vx - dy = -s$ and $0 \leq x < d$. Using $qv = pd + 1$,

$$\frac{qs}{d} = qy - \frac{qv}{d}x = qy - px - \frac{x}{d} = -(px - qy) - \frac{x}{d}.$$

Thus $\lceil qs/d \rceil = -(px - qy)$. Adding (u, c) raises the residual by one and keeps the first coordinate below $u + d = q$, which gives [equation \(15\)](#).

It remains to verify the two index partitions. We give the positive one directly. Both index lists are strictly increasing. For $1 \leq n < p$, the number of left indices at most n is

$$\left\lfloor \frac{cn}{p} \right\rfloor + 1,$$

while the number of right indices at most n is

$$\left\lceil \frac{vn}{p} \right\rceil = n - \left\lfloor \frac{cn}{p} \right\rfloor,$$

because $v = p - c$ and cn/p is nonintegral. Their sum is $n + 1$, exactly the number of positions in $\{0, \dots, n\}$. The endpoint counts at $n = 0, p$ are immediate, so every position $0, \dots, p$ occurs exactly once. The negative partition is identical with $q = u + d$. These index sets are the familiar finite rational Beatty partition; the vector labels supplied by [equations \(12\) to \(15\)](#) are the completion-specific content. $\square$

Remark 5.4 (Vector-valued Beatty shuffle). Theorem [5.3](#) says more than set inclusion. The entire residual order is reconstructed: the left parent populates the positions I_L^+ unchanged, while the translated right parent populates the complementary positions I_R^+ . The same holds on the negative side. The endpoint $\mu_{v,d}(v) = (1, 0)$ is omitted from [equation \(13\)](#) because the child endpoint $\mu_{p,q}(p) = (1, 0)$ is already inherited through the left parent.

Corollary 5.5 (Local–global Farey correspondence). *For a proper pair $p, q > 1$, the oriented Farey parents determine both the local jump law and the global ordered recursion. Conversely, the local positive jump set recovers the oriented parents and hence determines the ordered parent recursion.*

Proof. Forward determination is [theorems 4.1 and 5.3](#); reverse determination is [theorem 5.2](#). $\square$

6 The global Farey completion family

The preceding results concern a single Farey mediant. Iteration over the Stern–Brocot tree produces one recursively coherent completion family.

Definition 6.1 (Farey completion tree). Decorate every coprime node q/p of the Stern–Brocot tree by the pair of ordered profiles

$$\mathfrak{C}(p, q) = (\mathcal{M}_{p,q}^+, \mathcal{M}_{p,q}^-).$$

We call the resulting decorated tree the *Farey completion tree*.

Theorem 6.2 (Recursive coherence). *Whenever*

$$(p, q) = (c, u) + (v, d)$$

is a Farey mediant, both ordered profiles in $\mathfrak{C}(p, q)$ are determined uniquely from the two parent profiles by [theorem 5.3](#). Consequently the family

$$\{\mathfrak{C}(p, q) : p, q \geq 1, \gcd(p, q) = 1\}$$

is recursively coherent under Farey mediants.

Proof. Apply [theorem 5.3](#) at every proper interior node and stop when $p = 1$ or $q = 1$, where [theorem 3.4](#) supplies the explicit boundary profiles. The two index families partition every proper child profile, so the recursion determines every child entry exactly once. $\square$

Remark 6.3 (Classical tree, new decoration). The Stern–Brocot tree and its continued-fraction complexity are classical. The point of [theorems 6.1](#) and [6.2](#) is not a new rational tree, but a new vector-valued decoration: the shortest-completion profiles form a single parent-generated system rather than an unrelated collection indexed by $p : q$.

7 Scalar projection and universal spectra

For the Farey-coordinate formulas in the first part of this section assume $p, q > 1$; the boundary families are given by [theorem 3.4](#). Set

$$m = p + q, \quad A = c + u, \quad B = v + d.$$

Then $A + B = m$. From [equation \(6\)](#),

$$pA - mc = 1, \quad qB - md = 1. \tag{16}$$

Thus $A \equiv p^{-1} \pmod{m}$ and $B \equiv q^{-1} \pmod{m}$.

Proposition 7.1 (Scalar Farey profiles). *For $0 \leq r \leq p$,*

$$L_{p,q}(r) = Ar - m \left\lfloor \frac{cr}{p} \right\rfloor. \tag{17}$$

For $0 \leq s \leq q$,

$$L_{p,q}(-s) = Bs - m \left\lfloor \frac{ds}{q} \right\rfloor. \tag{18}$$

Proof. Take coordinate sums in [equations \(10\)](#) and [\(11\)](#). $\square$

Corollary 7.2 (Scalar jump law and affine periodicity). *For $0 \leq r < p$,*

$$L(r+1) - L(r) \in \{A, -B\},$$

with multiplicities v and c . Dually,

$$L(-(s+1)) - L(-s) \in \{B, -A\},$$

with multiplicities u and d . Moreover,

$$L(r+p) = L(r) + 1 \quad (r \geq 0) \tag{19}$$

and

$$L(-(s+q)) = L(-s) + 1 \quad (s \geq 0). \tag{20}$$

Proof. The jump statements are the ℓ^1 projections of [theorem 5.1](#). Their net positive drift is

$$vA - cB = v(c + u) - c(v + d) = uv - cd = 1,$$

and the negative drift is $uB - dA = 1$. Alternatively, [equations \(19\)](#) and [\(20\)](#) follow directly from the canonical strips by adding $(1, 0)$ or $(0, 1)$. $\square$

7.1 Fundamental spectra

Define

$$S_+(p, q) = \{L(r) : 0 \leq r < p\}, \quad S_-(p, q) = \{L(-s) : 0 \leq s < q\}.$$

Proposition 7.3 (Completion–Beatty identification). *As sets,*

$$S_+(p, q) = \left\{ \left\lceil \frac{mj}{p} \right\rceil : 0 \leq j < p \right\}, \quad (21)$$

and

$$S_-(p, q) = \left\{ \left\lceil \frac{mj}{q} \right\rceil : 0 \leq j < q \right\}. \quad (22)$$

Proof. For $0 \leq r < p$, the positive canonical coordinate y_r runs through every residue $j \in \{0, \dots, p-1\}$ exactly once. For fixed j , the least nonnegative first coordinate satisfying $px - qj \geq 0$ is $\lceil qj/p \rceil$, and therefore the corresponding completion length is

$$j + \left\lceil \frac{qj}{p} \right\rceil = \left\lceil \frac{(p+q)j}{p} \right\rceil.$$

The negative case is dual. $\square$

Theorem 7.4 (Universal Fundamental Spectrum). *For every coprime pair p, q with $m = p + q$,*

$$S_+(p, q) \uplus S_-(p, q) = \{0, 0, 2, 3, \dots, m-1\} \quad (23)$$

as multisets.

Proof. By [theorem 7.3](#), it is enough to show that the two nonzero ceiling sets partition $\{2, \dots, m-1\}$. For $1 \leq \ell \leq m$, define

$$\chi_p(\ell) = \left\lfloor \frac{p\ell}{m} \right\rfloor - \left\lfloor \frac{p(\ell-1)}{m} \right\rfloor, \quad \chi_q(\ell) = \left\lfloor \frac{q\ell}{m} \right\rfloor - \left\lfloor \frac{q(\ell-1)}{m} \right\rfloor.$$

Each quantity is 0 or 1. The condition $\chi_p(\ell) = 1$ is equivalent to $\ell = \lceil mj/p \rceil$ for some $1 \leq j \leq p-1$ when $2 \leq \ell \leq m-1$, and similarly for q . Since $q = m - p$ and $\gcd(p, m) = 1$, for every $2 \leq \ell \leq m-1$ neither $p\ell/m$ nor $p(\ell-1)/m$ is integral; using $\lfloor \ell - x \rfloor = \ell - \lceil x \rceil$ gives

$$\chi_p(\ell) + \chi_q(\ell) = 1.$$

Thus every integer $2, \dots, m-1$ belongs to exactly one of the two nonzero spectra. At $\ell = 1$ neither occurs, while both spectra contain 0. This proves [equation \(23\)](#). The resulting partition is the finite rational Beatty complementarity seen in [theorem 7.3](#). $\square$

7.2 Universal shell spectra

For $k \geq 0$, define

$$\Sigma_k^+ = \{L(kp + r) : 0 \leq r < p\}, \quad \Sigma_k^- = \{L(-kq - s) : 0 \leq s < q\}.$$

Theorem 7.5 (Universal Shell Spectrum). *For every $k \geq 0$,*

$$\Sigma_k^+ \uplus \Sigma_k^- = \{k, k, k+2, k+3, \dots, k+m-1\}. \quad (24)$$

Proof. By equations (19) and (20),

$$L(kp + r) = k + L(r), \quad L(-kq - s) = k + L(-s).$$

Translate theorem 7.4 by k . □

Corollary 7.6 (Shell generating polynomial). *If*

$$C_k(z) = \sum_{r=0}^{p-1} z^{L(kp+r)} + \sum_{s=0}^{q-1} z^{L(-kq-s)},$$

then

$$C_k(z) = z^k (2 + z^2 + z^3 + \dots + z^{m-1}),$$

and hence

$$\sum_{k \geq 0} C_k(z) t^k = \frac{2 + z^2 + z^3 + \dots + z^{m-1}}{1 - tz}.$$

Corollary 7.7 (Universal moment hierarchy). *For every integer $j \geq 1$,*

$$M_j(k) := \sum_{\ell \in \Sigma_k^+ \uplus \Sigma_k^-} \ell^j = 2k^j + \sum_{\ell=2}^{m-1} (k + \ell)^j.$$

In particular,

$$M_j(0) = \sum_{\ell=2}^{m-1} \ell^j.$$

Thus all combined shell power moments depend only on $m = p + q$.

7.3 Reflection symmetry

Write the nonzero positive spectrum in increasing order as

$$\ell_j^+ = \left\lceil \frac{mj}{p} \right\rceil, \quad 1 \leq j < p,$$

and define ℓ_j^- analogously with q .

Proposition 7.8 (One-sided reflection). *For $1 \leq j < p$,*

$$\ell_j^+ + \ell_{p-j}^+ = m + 1,$$

and for $1 \leq j < q$,

$$\ell_j^- + \ell_{q-j}^- = m + 1.$$

Consequently, with $h = (m + 1)/2$, every odd centered one-sided moment vanishes.

Proof. Because $\gcd(m, p) = 1$, mj/p is nonintegral for $1 \leq j < p$. Hence

$$\left\lceil \frac{m(p-j)}{p} \right\rceil = m - \left\lfloor \frac{mj}{p} \right\rfloor,$$

which sums with $\lceil mj/p \rceil$ to $m + 1$. The negative identity is dual, and centered odd powers cancel in symmetric pairs. $\square$

8 Classical arithmetic interfaces

The results through [section 7](#) were obtained from shortest-completion minimization, canonical kernel reduction, and Farey unimodularity. We now record how classical arithmetic structures appear as projections. The purpose of this section is interpretive rather than foundational.

8.1 Mechanical and Christoffel coding

The positive wrap indicator is

$$\varepsilon_r = \left\lfloor \frac{c(r+1)}{p} \right\rfloor - \left\lfloor \frac{cr}{p} \right\rfloor \in \{0, 1\}. \quad (25)$$

Then

$$\mu(r+1) - \mu(r) = (u, c) - \varepsilon_r(q, p).$$

Thus $\varepsilon_r = 0$ selects (u, c) and $\varepsilon_r = 1$ selects $(-d, -v)$. The finite binary sequence $(\varepsilon_r)_{r=0}^{p-1}$ is a rational lower mechanical word; under the standard finite conventions it is a Christoffel encoding, up to the usual choice of slope orientation and alphabet. Standard factorization, continued fractions, and the relation with the Stern–Brocot tree are classical; see [\[2\]](#).

The completion interpretation is that the mechanical/Christoffel coding records exactly when canonical kernel reduction occurs. This should not be read as a novelty claim for the coding itself.

8.2 Unimodular realization and the distinction from standard factorization

Let

$$k_r = \left\lfloor \frac{cr}{p} \right\rfloor \quad (0 \leq r \leq p)$$

and define

$$J_+ = \begin{pmatrix} u & -d \\ c & -v \end{pmatrix}.$$

Since $uv - cd = 1$, one has $\det J_+ = -1$. Using $(q, p) = (u, c) - (-d, -v)$ in [equation \(10\)](#) gives

$$\mu_{p,q}(r) = (r - k_r)(u, c) + k_r(-d, -v) = J_+ \begin{pmatrix} r - k_r \\ k_r \end{pmatrix}. \quad (26)$$

Therefore

$$J_+^{-1} \mu_{p,q}(r) = \begin{pmatrix} r - \lfloor cr/p \rfloor \\ \lfloor cr/p \rfloor \end{pmatrix},$$

which is the vertex sequence of a rational mechanical lattice path with v steps of one type and c of the other. Thus the local completion walk is a unimodular image of classical Christoffel/mechanical geometry. This is consistent with digital-straight-line theory, where arithmetic remainders, minimal

$a + b$ parameters, Bézout vectors and Stern–Brocot slope updates already occur together [4], and with classical Christoffel duality and modular-inverse phenomena [3].

It is important, however, not to identify [theorem 5.3](#) with standard Christoffel factorization. Standard factorization decomposes a proper Christoffel word into a *contiguous concatenation* of two Christoffel factors associated with the Farey parents [2]. By contrast, [theorem 5.3](#) compares three different canonical minimizer maps,

$$\mu_{p,q}, \quad \mu_{c,u}, \quad \mu_{v,d},$$

and partitions the child residual positions into two *interleaved* sets

$$I_L^+ = \left\{ \left\lfloor \frac{pr}{c} \right\rfloor : 0 \leq r \leq c \right\}, \quad I_R^+ = \left\{ \left\lfloor \frac{ps}{v} \right\rfloor + 1 : 0 \leq s < v \right\}.$$

The corresponding child-vector subsequences are exactly the left-parent completion profile and a fixed translate of the right-parent completion profile. Hence standard factorization explains why Farey parents and Christoffel structure are nearby, but it is not the same residual-indexed cross-parameter identity.

The two scalar jump amplitudes are

$$A = c + u, \quad B = v + d,$$

so, under the standard Christoffel factorization conventions, the corresponding factor lengths also appear as the ℓ^1 projections of the two completion unit lifts.

8.3 Dedekind structure of the one-sided second moment

The combined spectrum is universal, but its split into positive and negative parts depends on $p : q$. This ratio dependence is already visible in the centered second moment.

Let

$$((x)) = \begin{cases} \{x\} - \frac{1}{2}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}, \end{cases}$$

and define the classical Dedekind sum

$$s(h, k) = \sum_{j=1}^{k-1} \left(\left(\frac{j}{k} \right) \right) \left(\left(\frac{hj}{k} \right) \right).$$

Put $h_0 = (m + 1)/2$. For the positive sorted spectrum,

$$\ell_j^+ - h_0 = m \left(\left(\frac{j}{p} \right) \right) - \left(\left(\frac{qj}{p} \right) \right). \quad (27)$$

Since multiplication by q permutes the nonzero residue classes modulo p ,

$$\sum_{j=1}^{p-1} \left(\left(\frac{qj}{p} \right) \right)^2 = \sum_{j=1}^{p-1} \left(\left(\frac{j}{p} \right) \right)^2 = \frac{(p-1)(p-2)}{12p}.$$

Squaring [equation \(27\)](#) therefore gives

$$D_2^+(p, q) := \sum_{r=1}^{p-1} \left(L(r) - \frac{m+1}{2} \right)^2 = \frac{(m^2 + 1)(p-1)(p-2)}{12p} - 2m s(q, p). \quad (28)$$

Similarly,

$$D_2^-(p, q) = \frac{(m^2 + 1)(q - 1)(q - 2)}{12q} - 2m s(p, q). \quad (29)$$

By [theorem 7.4](#),

$$D_2^+(p, q) + D_2^-(p, q) = \frac{(m - 1)(m - 2)(m - 3)}{12}. \quad (30)$$

Substituting [equations \(28\)](#) and [\(29\)](#) into [equation \(30\)](#) recovers the classical Dedekind reciprocity law

$$s(q, p) + s(p, q) = -\frac{1}{4} + \frac{1}{12} \left(\frac{q}{p} + \frac{p}{q} + \frac{1}{pq} \right);$$

see [\[6\]](#). Dedekind reciprocity is not a new result here. Its completion interpretation is that Dedekind sums quantify the ratio-dependent one-sided splitting of an otherwise universal completion spectrum.

Remark 8.1 (Higher even moments). By [theorem 7.8](#), all odd centered moments vanish. Higher even moments expand into products of periodic Bernoulli functions and hence into generalized Dedekind-type sums. Since such sums and their reciprocity laws have an extensive independent literature, including [\[11\]](#), we do not pursue new reciprocity statements here.

9 Computational discovery and validation

The results above are proved symbolically and do not depend on computation. Numerical experimentation served first as a discovery device and later as an independent audit of signs, endpoints, and floor/ceiling conventions.

An initial audit covered every coprime ordered pair $2 \leq p, q \leq 100$. A second reviewer-oriented audit extended the exhaustive range to

$$2 \leq p, q \leq 200,$$

covering 24,064 coprime ordered pairs, and added 30,000 randomized coprime pairs with $2 \leq p, q \leq 5000$. Shortest-completion vectors were constructed independently from [theorem 3.2](#). For every residual in the positive and negative fundamental ranges, these direct solutions were compared with the formulas of [sections 4](#) and [5](#). The audits verified, without exception:

- the positive and negative canonical unit-lift laws;
- the closed vector formulas [equations \(10\)](#) and [\(11\)](#);
- all four ordered parent-to-median identities [equations \(12\)](#) to [\(15\)](#);
- the corresponding positive and negative index partitions.

No counterexample was found in either the exhaustive or randomized vector audits.

A separate scalar audit tested every coprime ordered pair

$$1 \leq p, q \leq 150,$$

comprising 13,715 ratios. For each pair, the fundamental spectrum and the translated shells $k = 0, 1, \dots, 7$ were generated directly from shortest completions. In every case,

$$\Sigma_k^+ \uplus \Sigma_k^- = \{k, k, k + 2, k + 3, \dots, k + p + q - 1\}.$$

These computations are not proof premises. The direct completion oracle used only the defining equation $px - qy = r$ and the canonical-strip bounds, rather than the closed Farey formulas being tested. The experiments therefore function only as independent consistency checks.

10 Conclusion

We have studied the ordered family of shortest nonnegative integer solutions of

$$px - qy = r,$$

with p and q coprime, motivated by shortest completion in rational balance languages.

For a proper Farey decomposition

$$(p, q) = (c, u) + (v, d), \quad uv - cd = 1,$$

the positive shortest-completion profile advances through two kernel-equivalent unit-residual lifts,

$$(u, c) \quad \text{and} \quad (-d, -v).$$

Canonical reduction in the strip $0 \leq y < p$ selects between them, and the negative profile has the corresponding dual dynamics. After a unimodular coordinate change this local walk is a classical rational mechanical path, so its Christoffel form is best viewed as the local arithmetic mechanism rather than as the principal novelty.

The principal completion-specific result is the cross-parameter Ordered Farey Recursion Theorem. It reconstructs the entire median minimizer section from the two parent minimizer sections through complementary floor/ceiling index maps: the two parent profiles appear as interleaved child subsequences, one after a fixed translation. The local jump vectors recover the oriented parents, and iteration yields a recursively coherent family over the Stern–Brocot tree.

Returning to the motivating automaton, the map $r \mapsto \mu_{2,1}(r)$ is exactly the shortest-demand semantics realized by its distributed state–stack representation; the present results show that this semantic object belongs to a much larger Farey-recursive arithmetic family.

Scalar projection produces a complementary universal phenomenon. Writing $m = p + q$, the combined positive and negative fundamental spectrum is always

$$\{0, 0, 2, 3, \dots, m - 1\},$$

and every residual shell is an affine translate of this multiset. Hence all combined shell moments depend only on m . Classical Beatty, mechanical/Christoffel, and Dedekind structures arise as arithmetic shadows of this vector theory rather than as proof premises.

Taken together, these results organize the coprime rational balance family as a single Farey-recursive system of shortest-completion profiles, with locally recoverable parent data and universal scalar spectra.

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